Project: **BODO REMEDIATION ANALYTICS - 2025 LAND USE/LAND COVER AND CHANGE DETECTION STUDIES**

INTRODUCTION:

The ongoing Bodo project involves Shoreline Clean-up and Assessment Technique (SCAT) activities including onsite verification of remediation works, field sampling/laboratory analyses and mangrove revegetation activities, all running concurrently. To complement field observations, it was decided that a time lapse analysis of satellite imagery should also be applied for the monitoring of mangrove recovery within the area of interest.

This report contains the statistical analysis result for the land use/land cover classification of the satellite imagery over the Bodo remediation area of interest acquired on the 18th of December 2024 (For 2025 study year report). The result highlights the progress of the OGR team on the mangrove regrowth and quantifies the area of the vegetation cover as at the time of satellite data capture. The Area of interest (AOI) was advised by the ORP team and has been consistently used for all previous and present studies.

STUDY OBJECTIVES:

The objectives of this study are:

- To identify, map, extract and evaluate area covered by vegetation.
- To identify evidence of vegetation regrowth within Bodo AOI and determine the differences in the status of land use and land cover over 2 epochs within the Bodo AOI from January 2024 to December 2024 (for January 2025 study).
- To determine the different variations and estimate the progress of mangrove regrowth within the Bodo AOI.

The methodology to achieve these objectives include Image compositing and classification using the Support Vector Machine (SVM) algorithm & geometry extraction calculations and data analytics.

STUDY AREA:

The study area covers Bodo communities within the Oil Mining Lease (OML 11) in the Niger Delta. The total area of study is approximately 20.98 km^2.

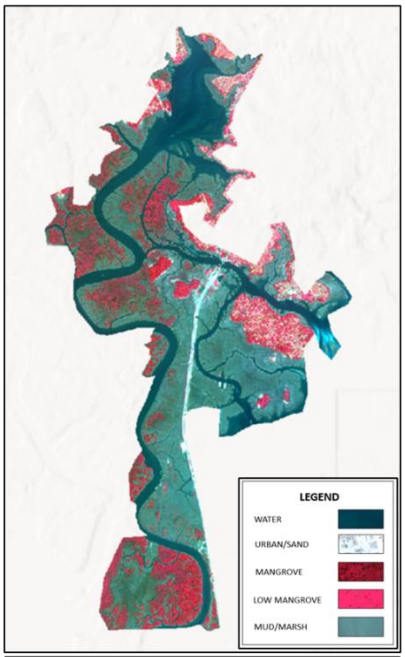
The terrain is mainly mangrove swamps with creeks and tidal or mudflats, the study area is susceptible to tidal actions. The study area is accessible via land and rivers, and the climate is tropical monsoon.

METHOLOGY AND RESULTS:

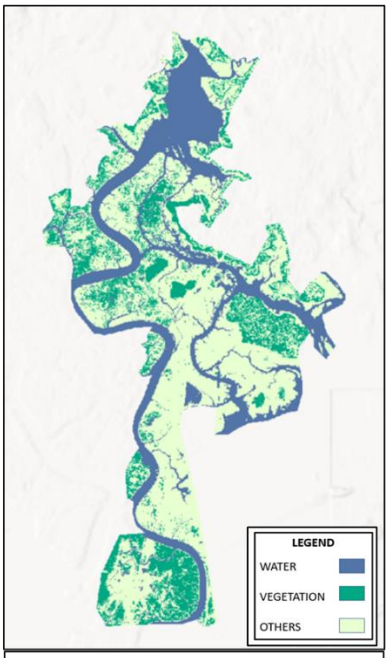
The satellite data was first corrected for the attenuating effect of atmospheric moisture, dust particles and fine clouds (so called atmospheric correction) and referencing/positioning errors. A 'False Colour' (FC) image was then generated from the corrected data i.e., the near-infrared reflection intensity of the image was visualized in red to highlight the chlorophyll cells in plant leaves which strongly absorb red EM radiation, but at the same time strongly reflect near-infrared wavelengths. The reddish hues gave an indication of vegetation of varying intensity and coverage, whereas blue, grey and white hues indicate water, mud/marsh and built-up areas (both categorize as Others in the December 2024 classification result).

The satellite image data (18th December 2024) was georeferenced with the January 2024 satellite image and classified using the support vector machine classification method from the classification wizard tool in ArcGIS Pro to generate land use and land cover class distribution for the area of interest. The baseline information for each class was extracted from the False Colour Composites (FC) combination of the RGB raster data and a statistical analysis of the identified feature classes was done to quantify the area for each land use/land cover class.

True Colour Composite 18th December 2024 (For 2025 Study)

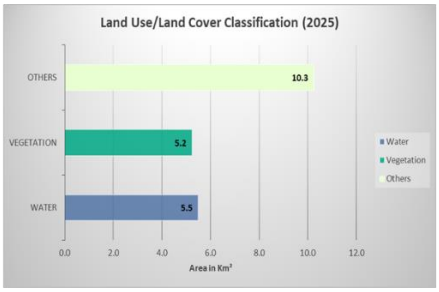


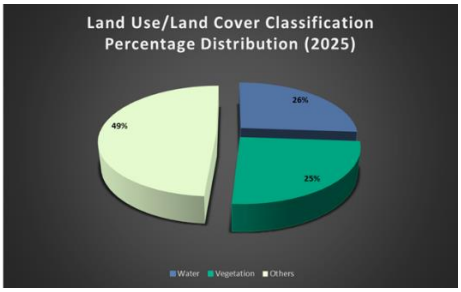
False Colour Composite 18th December 2024 (For 2025 Study)



Land use/Land cover Classification 18th December 2024 (For 2025 Study)

Epoch: Dec 2024 (For 2025 Study)		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.5	26%
Vegetation	5.2	25%
Others	10.3	49%
TOTAL	20.99	100%





LANDUSE AND LAND COVER ANALYSIS:

The False Colour Composite (FC) image clearly distinguished between the built-up areas and sand features (Roads and bare soil), and vegetation areas within the area of interest. The vegetated areas are clearly distinct in red, in comparison to their bare and partly submerged surroundings (i.e., mud/marsh areas). Shallow tidal areas and deeper channels of the creeks within the area of interest were also clearly distinguishable. The result shows that water covers 27.2%, vegetation covers 20.4% and others cover 52.4% of the entire AOI.

Water:

The water class reduced from 8.0 km^2 in January 2024 to 5.5 km^2 in December 2024. This reduction in water level may be attributed to the changes in tide and water levels as at the time of image acquisition.

Vegetation:

The vegetation class reduced from 6.3 km^2 in January 2024 to 5.2 km^2 (approximate reduction of 1.1 km^2) in December 2024 (for 2025 study year). This reduction is probably due to noticeable pockets and large mangrove vegetation loss distributed across the Bodo area of interest. From visual interpretation, the vegetation loss may be attributed to environmental/natural phenomena or human activities and interference. Judging from the image interpretation and analysis, the mangrove regrowth process by the ORP team appears to give a false progressive representation due to the significant vegetation loss observed across the entire Bodo AOI. We propose that all the Cell locations and the entire Bodo study area be given consideration for new replanting of mangrove seedlings most especially in areas where there are defoliation of vegetation and as well locations that the vegetation have completely disappeared as highlighted in the report (Pages 9 – 13).

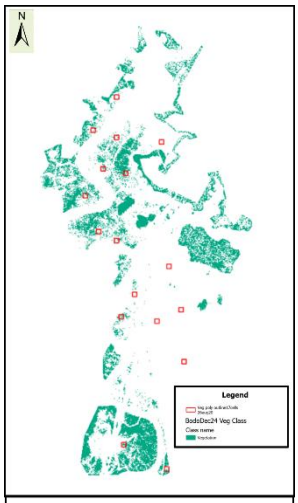
Others:

The others class increased from 6.6 km^2 in January 2024 to 10.3 km^2 in December 2024 (for 2025 study year). The reason for the increase is probably attributed to changes in the water levels due to low tide showing extended riverbank covering most of the mud flats area.

Comment:

Further analysis of the Bodo satellite imagery **shows the vegetation loss spreads across the Bodo study area in several pockets and locations (pages 9-13). The vegetation loss is most likely attributed to human or natural interference.**

CHANGE ANALYSIS FOR MANGROVE RECOVERY WITHIN BODO AOI (JAN 2024 - DEC 2024)

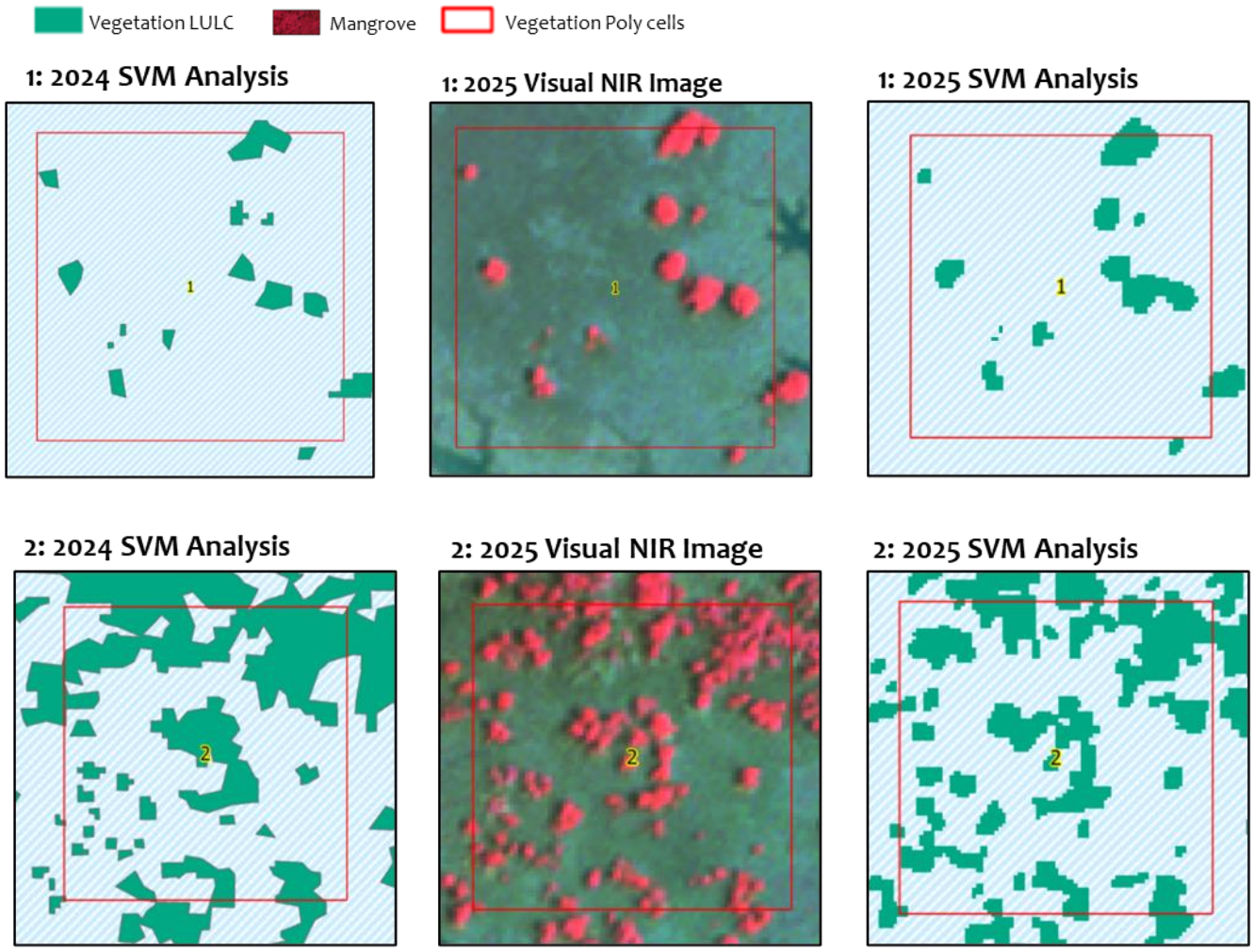


Clip showing extracted vegetation class and sample cell locations for Bodo Remediation Area of Interest

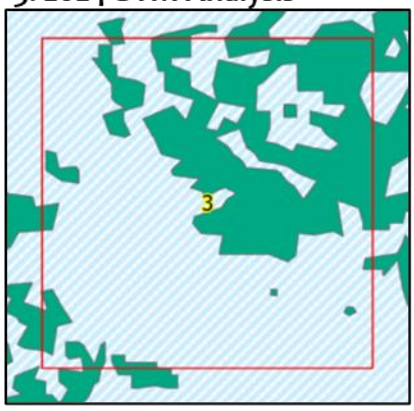
A change detection analysis was done between the two image scenes (Jan 2024 and Dec 2024). This is to help identify the evidence of vegetation recovery within the area of interest and to determine the differences in the status of land use and land cover within the Bodo study area.

Furthermore, to map out the variations and estimate the progress of the mangrove regrowth, a total number of 17 cell sites were identified to help measure each mangrove presence visually. These 17 cells were provided by the ORP team. These measurements were then used as a basis for comparing the image processing results for those areas. These 17 cell sample locations were selected to have various levels of mangrove coverage and were roughly distributed within the area of interest with a focus near the Bodo City. This principle was applied to the classification analysis over the Bodo remediation site (Jan 2024 – Dec 2024) to visually interpret and analyse the amount of vegetation regrowth and reduction within each vegetation sample cell.

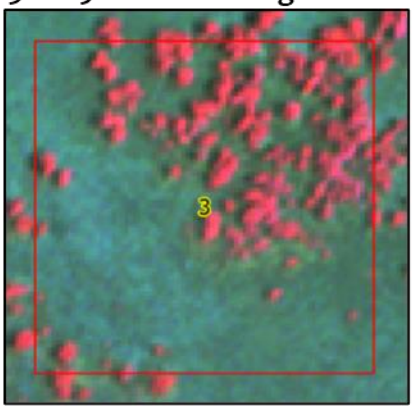
Below are the interpretation and classification results between the January 2024 SVM Analysis and the December 2024 SVM Analysis (representing 2025 year of study) using the 17cells vegetation polygon outline data integrated with the December 2024 Bodo NIR raster data. Each cell has been categories as 1 - 17. Cells 1 – 13 and 16 – 17 are within the Bodo cleanup AOI while Cells 14 and 15 are within the Ataba cleanup AOI. Margin of Error is + or – 5m.



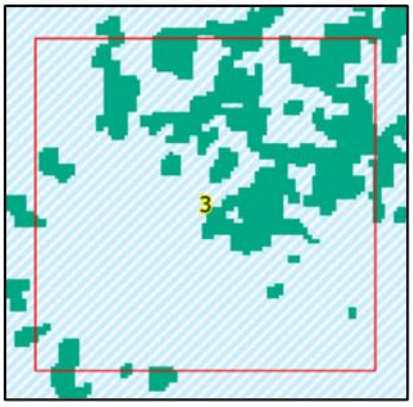
3: 2024 SVM Analysis



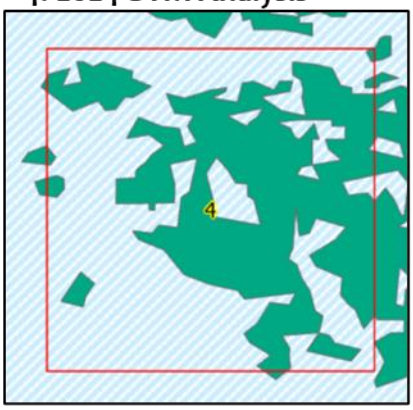
3: 2025 Visual NIR Image



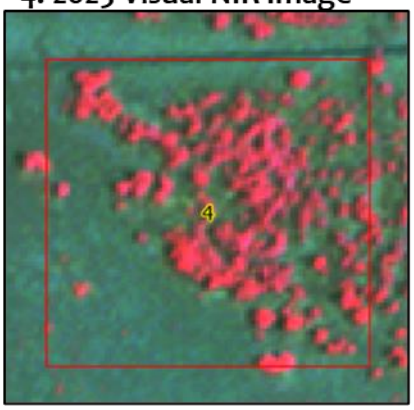
3: 2025 SVM Analysis



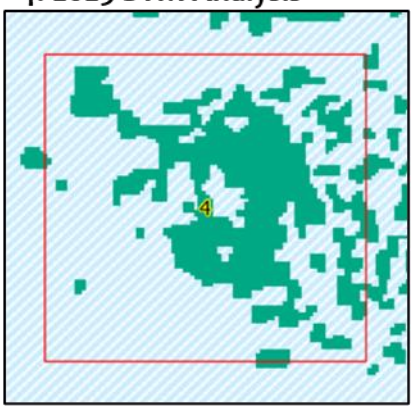
4: 2024 SVM Analysis



4: 2025 Visual NIR Image

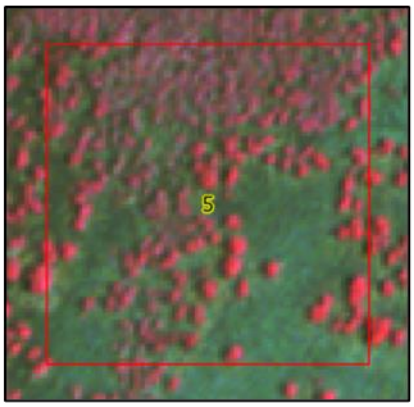


4: 2025 SVM Analysis

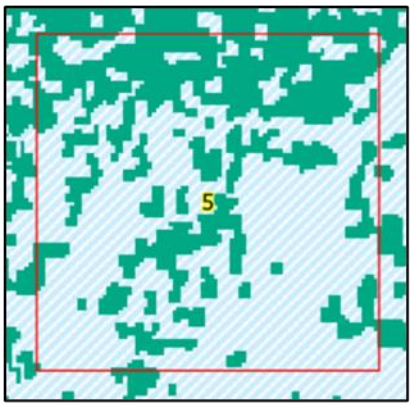


5: 2024 SVM Analysis

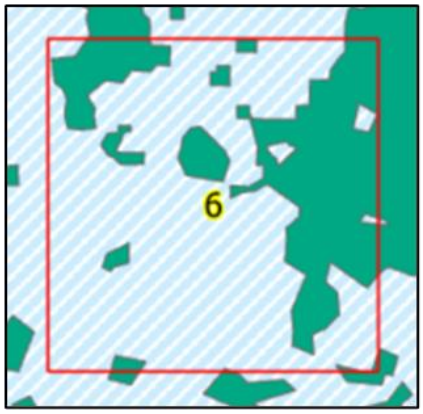
5: 2025 Visual NIR Image



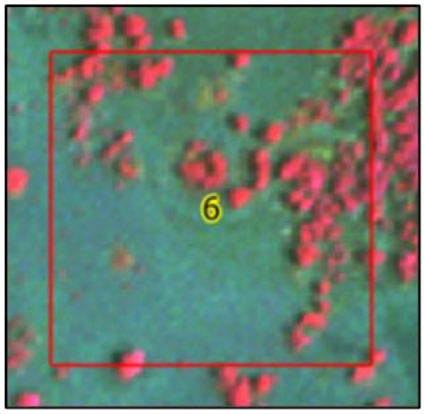
5: 2025 SVM Analysis



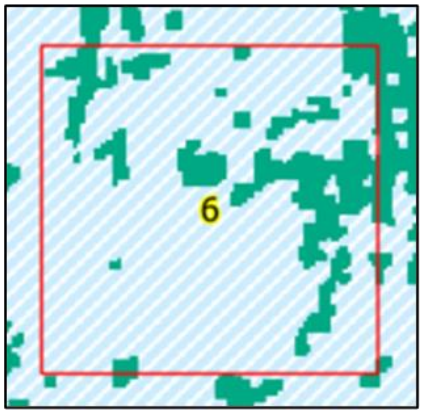
6: 2024 SVM Analysis



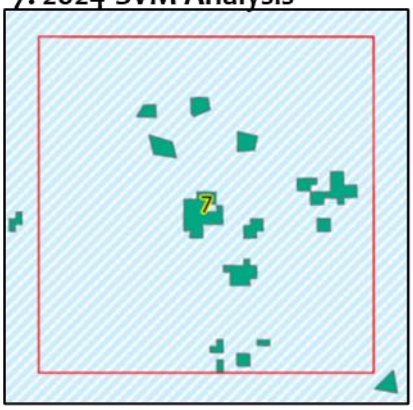
6: 2025 Visual NIR Image



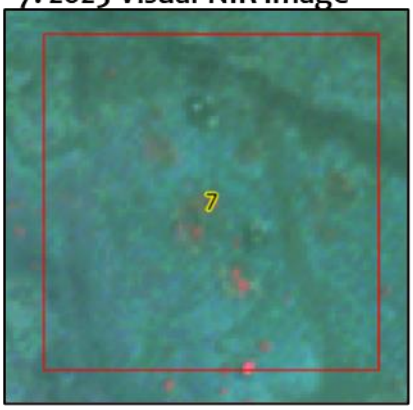
6: 2025 SVM Analysis



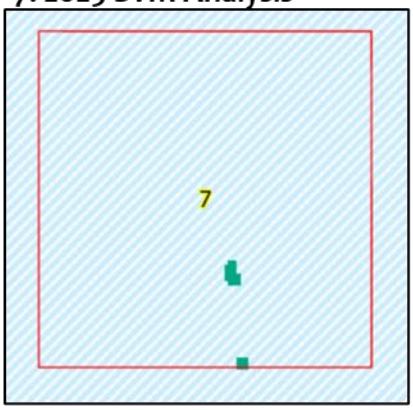
7: 2024 SVM Analysis



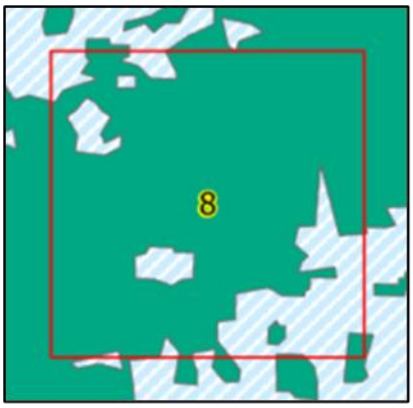
7: 2025 Visual NIR Image



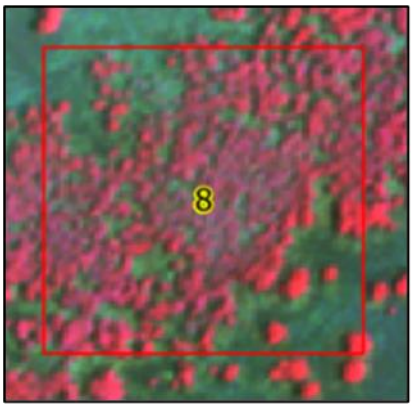
7: 2025 SVM Analysis



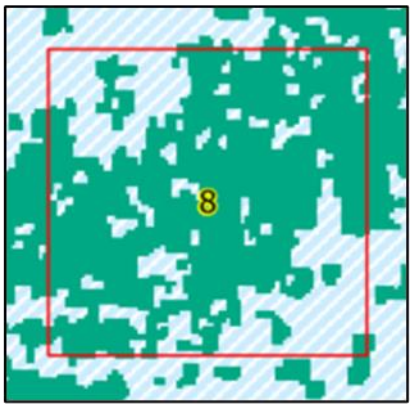
8: 2024 SVM Analysis



8: 2025 Visual NIR Image

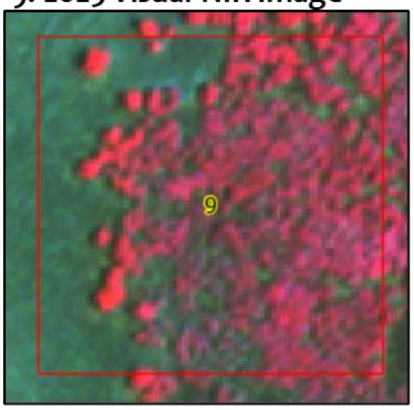


8: 2025 SVM Analysis

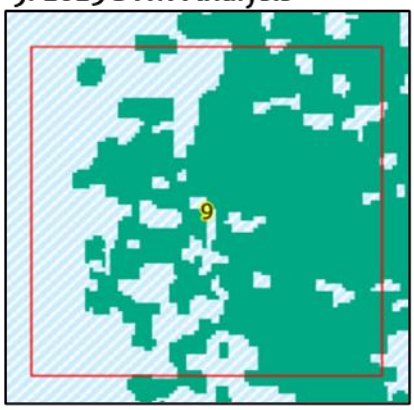


9: 2024 SVM Analysis

9: 2025 Visual NIR Image

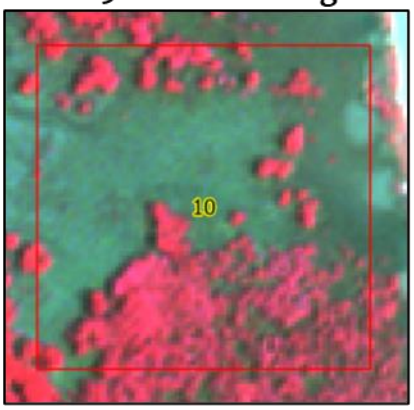


9: 2025 SVM Analysis

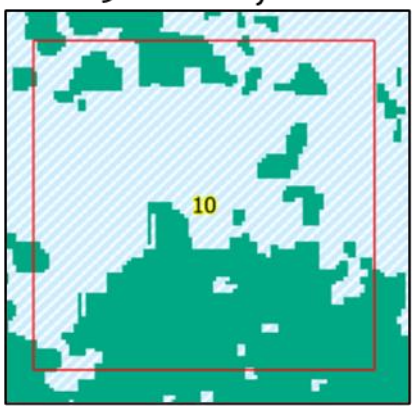


10: 2024 SVM Analysis

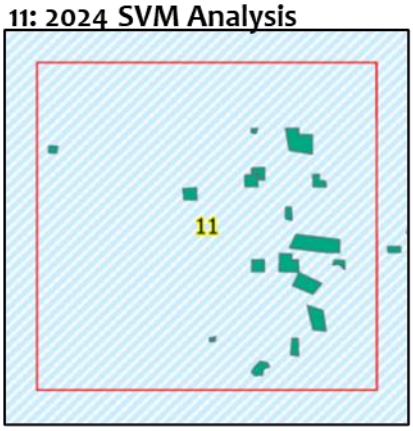
10: 2025 Visual NIR Image



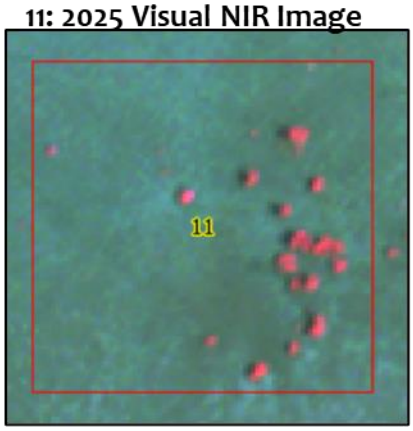
10: 2025 SVM Analysis



11: 2024 SVM Analysis



11: 2025 Visual NIR Image



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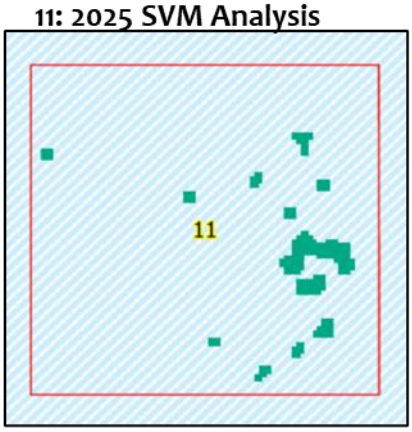
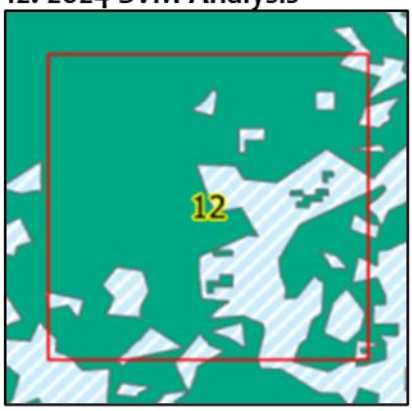
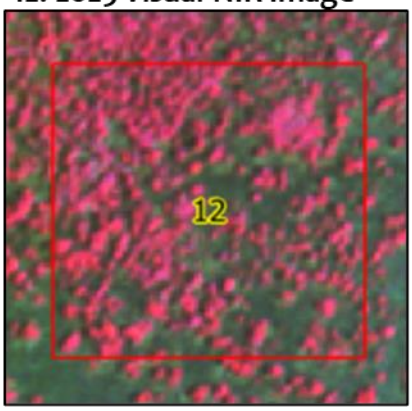


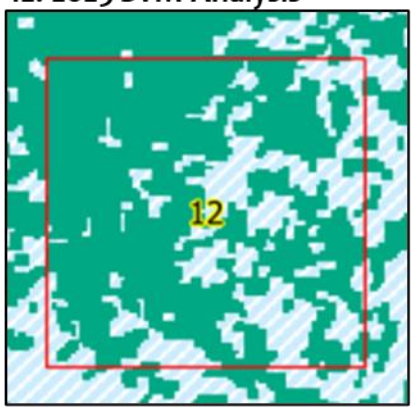
12: 2024 SVM Analysis



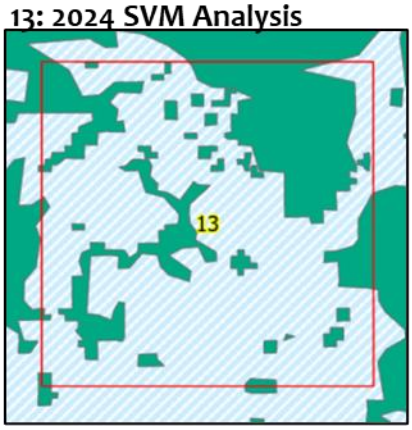
12: 2025 Visual NIR Image



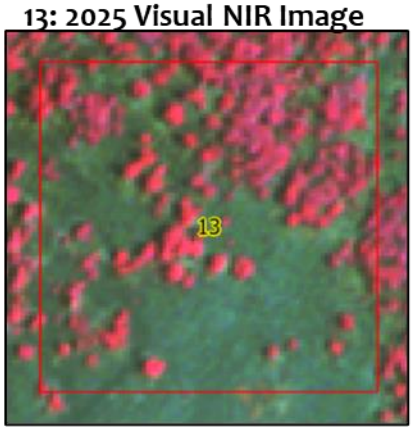
12: 2025 SVM Analysis



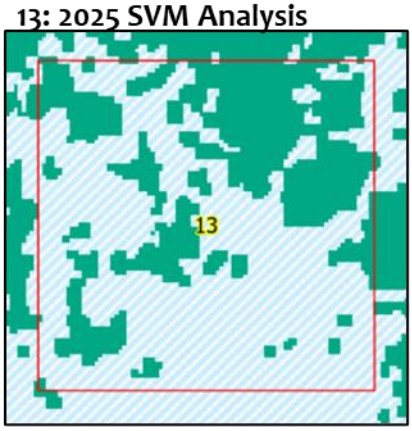
13: 2024 SVM Analysis



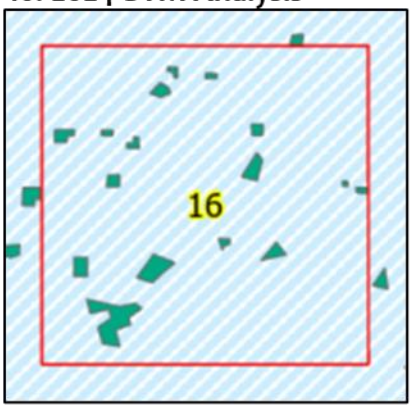
13: 2025 Visual NIR Image



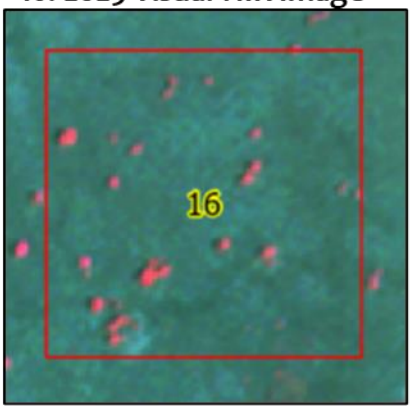
13: 2025 SVM Analysis



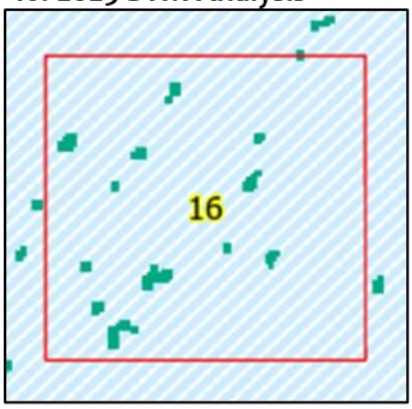
16: 2024 SVM Analysis



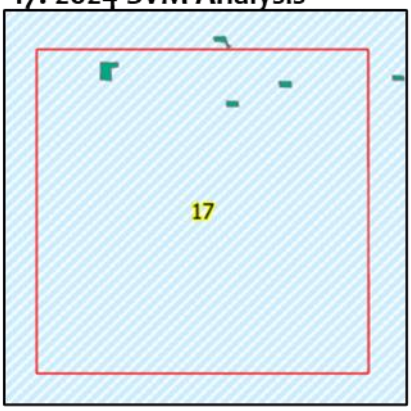
16: 2025 Visual NIR Image



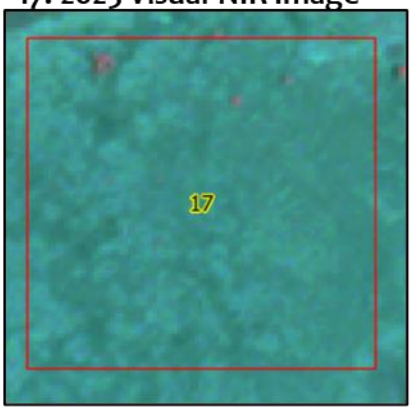
16: 2025 SVM Analysis



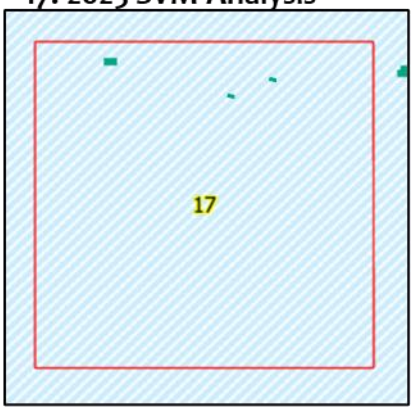
17: 2024 SVM Analysis



17: 2025 Visual NIR Image

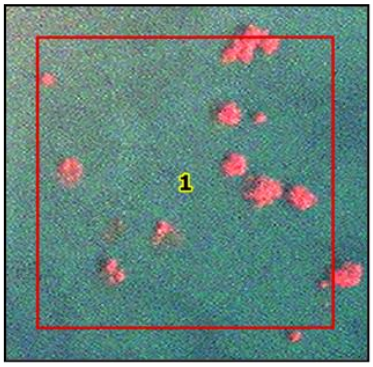


17: 2025 SVM Analysis

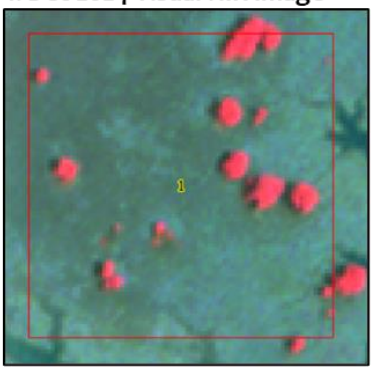


Change Detection Analysis Between January 2024 Visual NIR and December 2024 (For 2025 Study Year) Visual NIR Images (Cells 1 – 13, 16-17)

1: Jan 2024 Visual NIR Image

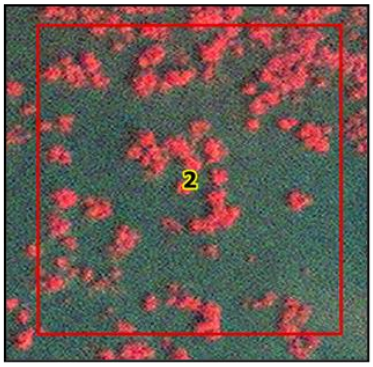


1: Dec 2024 Visual NIR Image

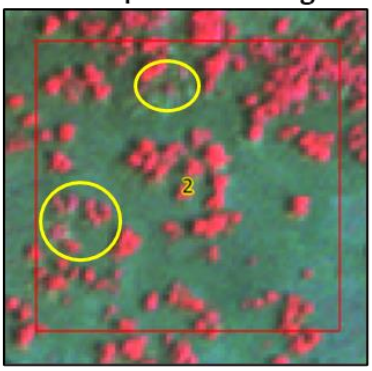


1 – No obvious vegetation change noticed between January 2024 – December 2024 period. Vegetation foliage appear healthy. Size of vegetation and area covered remains unchanged.

2: Jan 2024 Visual NIR Image

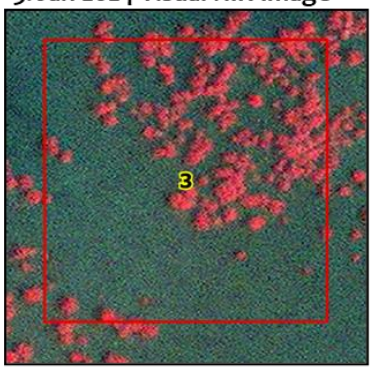


2: Dec 2024 Visual NIR Image

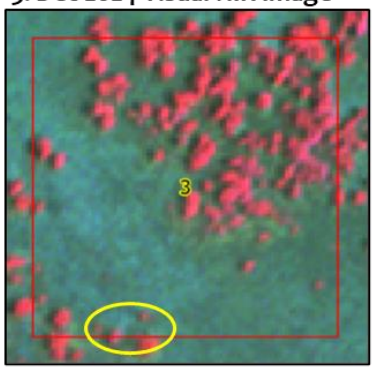


2 – The Pattern and distribution of vegetation appear similar. A noticeable reduction in vegetation foliage and size, circled in yellow, was observed when comparing both images. This is most likely due to human or natural interference.

3: Jan 2024 Visual NIR Image

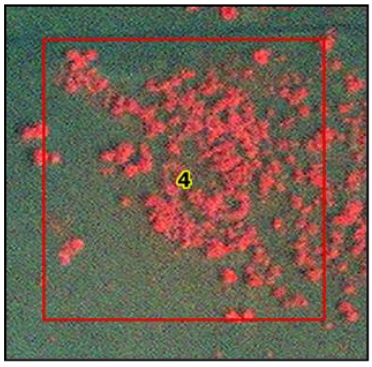


3: Dec 2024 Visual NIR Image

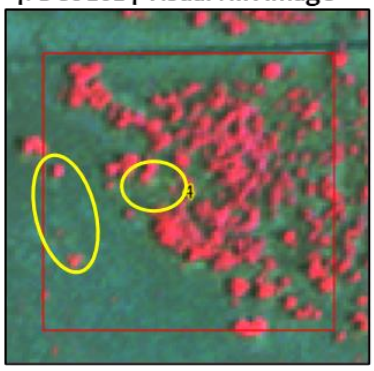


3 – The Pattern and distribution of vegetation appear similar. A small reduction in vegetation foliage and size, circled in yellow. Aside the tiny reduction, no obvious change noticed within Cell 3.

4: Jan 2024 Visual NIR

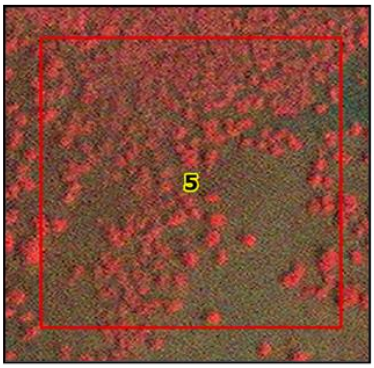


4: Dec 2024 Visual NIR Image

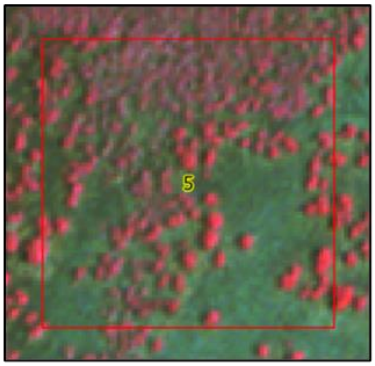


4 – Cell 4 has similar vegetation distribution between January 2024 and December 2024 period. There are no significant growth observed within the period of review. Loss of vegetation highlighted within the yellow circle.

5: Jan 2024 Visual NIR Image

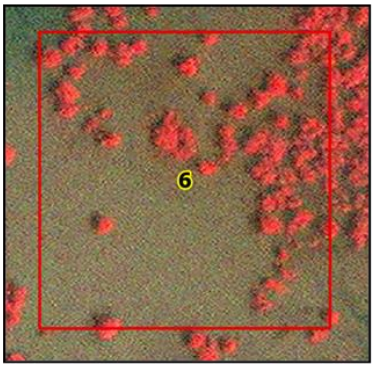


5: Dec 2024 Visual NIR Image

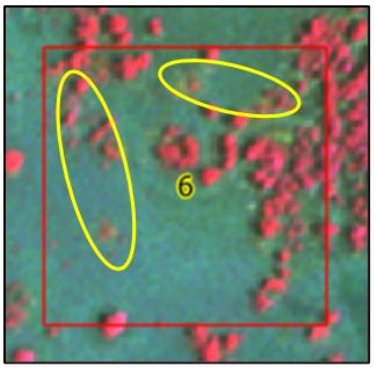


5 – The Pattern and distribution of vegetation appear similar There are no significant mangrove growth or difference observed.

6: Jan 2024 Visual NIR Image

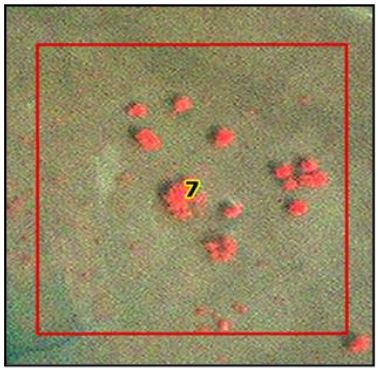


6: Dec 2024 Visual NIR Image

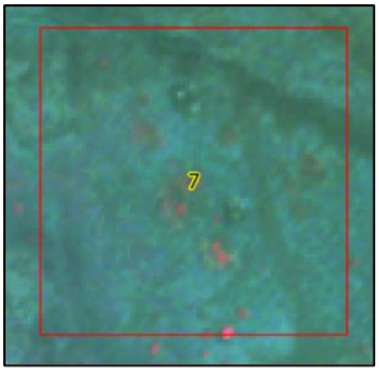


6 – The distribution and pattern of vegetation appear similar. A noticeable reduction in vegetation foliage and size, circled in yellow, was observed in the December 2024 image. This is most likely due to human or natural interference

7: Jan 2024 Visual NIR Image

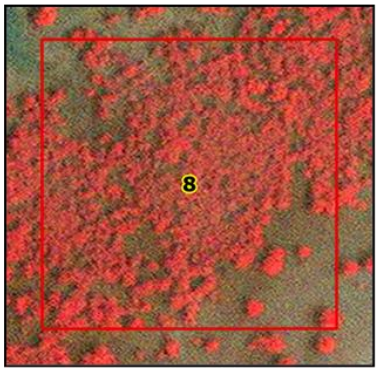


7: Dec 2024 Visual NIR Image

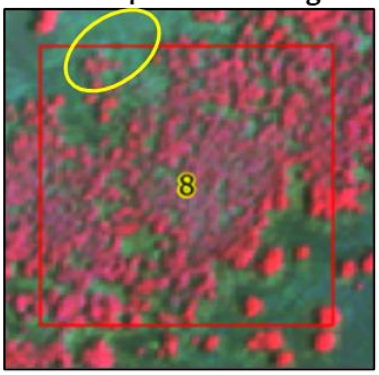


7 – The December 2024 Visual NIR image shows visible vegetation loss and complete disappearance of tree foliage. This is most likely due to human or natural interference.

8: Jan 2024 Visual NIR Image

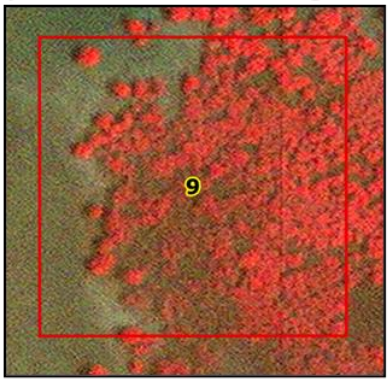


8: Dec 2024 Visual NIR Image

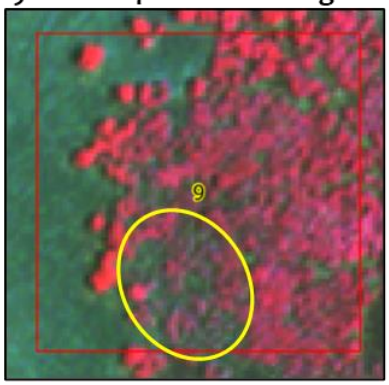


8 - The distribution and pattern of vegetation appear similar for both year of study. Loss of vegetation highlighted within the yellow circle.

9: Jan 2024 Visual NIR Image

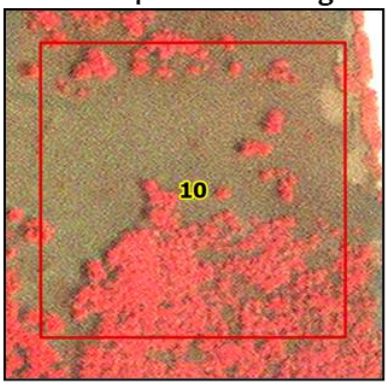


9: Dec 2024 Visual NIR Image

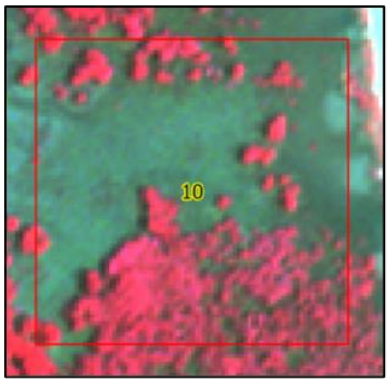


9 - The Pattern and distribution of vegetation appear similar between both Visual NIR images. As highlighted in January 2024, the defoliation of the vegetation, circled in yellow, is still very evident in the December 2024 image.

10: Jan 2024 Visual NIR Image

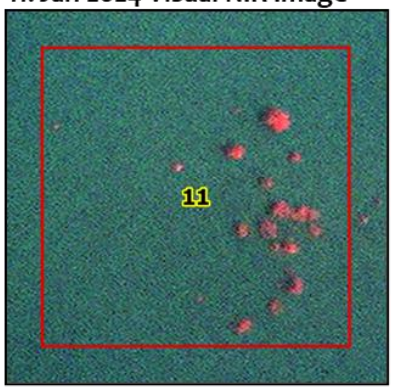


10: Dec 2024 Visual NIR Image

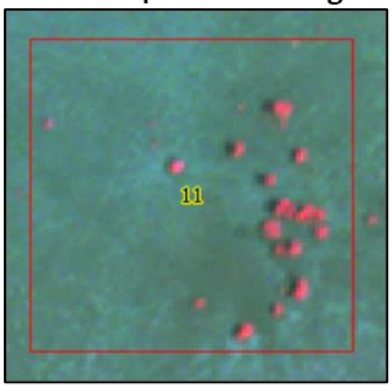


10 – Similar distribution of vegetation and tree foliage. No obvious mangrove growth addition or reduction

11: Jan 2024 Visual NIR Image

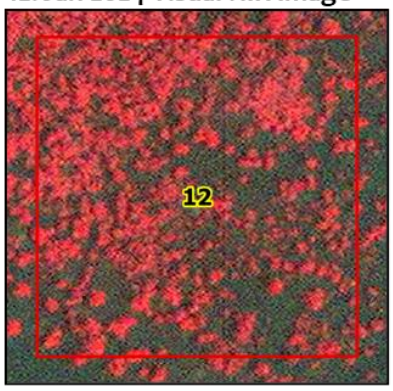


11: Dec 2024 Visual NIR Image

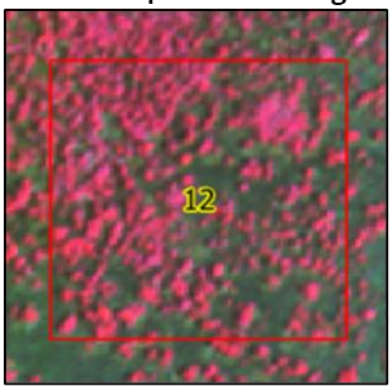


11 - The Pattern and distribution of vegetation appear similar between both Visual NIR images. There are no significant mangrove growth or difference observed.

12: Jan 2024 Visual NIR Image

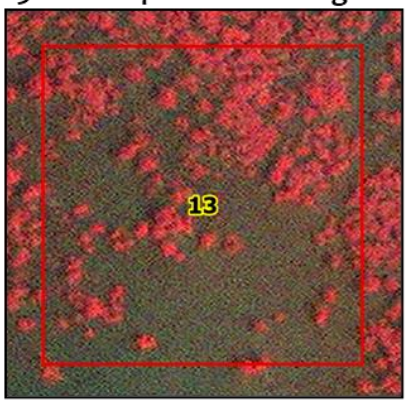


12: Dec 2024 Visual NIR Image

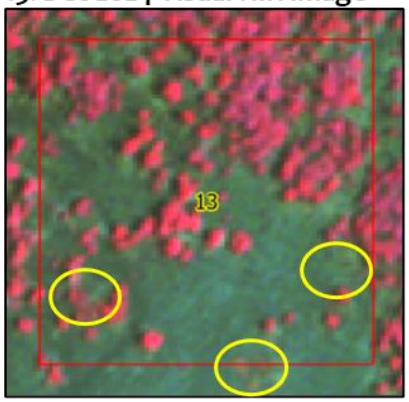


12 - The distribution and pattern of vegetation appear similar between both year of study. There are no significant growth or difference observed.

13: Jan 2024 Visual NIR Image

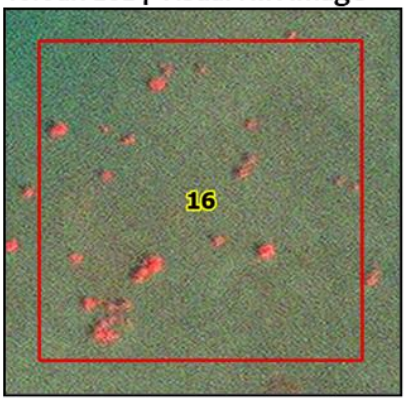


13: Dec 2024 Visual NIR Image

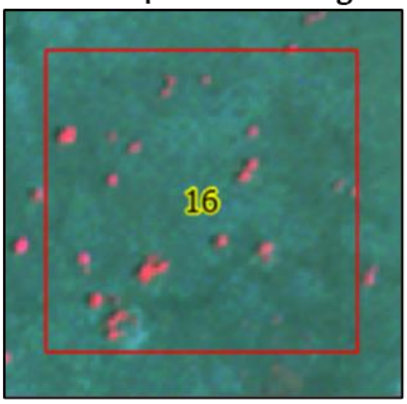


13 - The distribution and pattern of vegetation appear similar between both year of study. Some noticeable disappearance and reduction of vegetation foliage and size, circled in yellow, was observed in the December 2024 image.

16: Jan 2024 Visual NIR Image

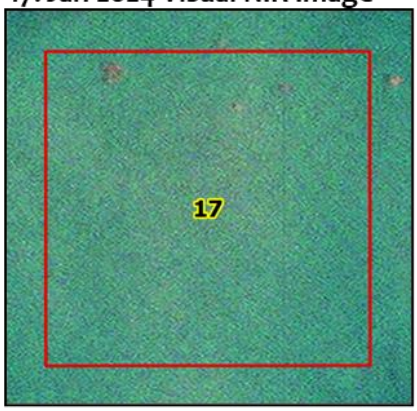


16: Dec 2024 Visual NIR Image

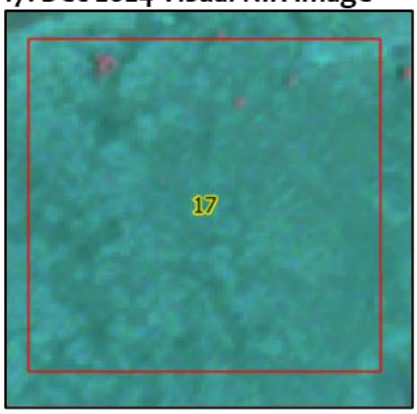


16 - There are no significant growth observed between both images. The distribution and pattern of vegetation appear similar between both year of study.

17: Jan 2024 Visual NIR Image

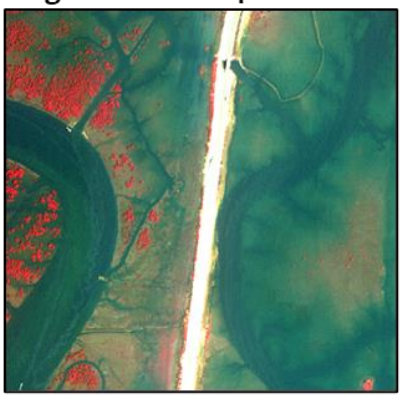


17: Dec 2024 Visual NIR Image

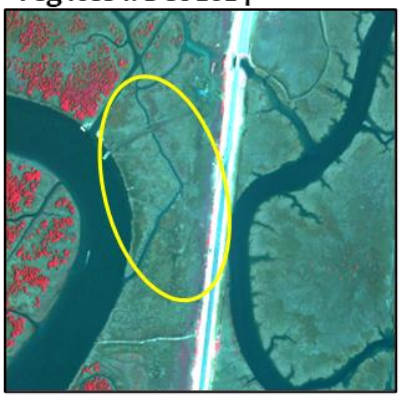


17 - There are no significant mangrove regrowth observed between both images. Cell 17 seems to be devoid of any vegetation regrowth since the last study.

Veg loss 1: Jan 2024

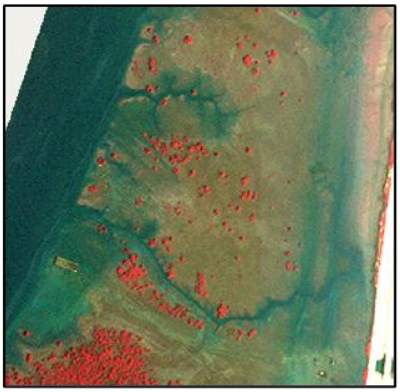


Veg loss 1: Dec 2024



Veg Loss 1 – Significant vegetation loss observed in area circled in yellow. This is most likely due to human or natural interference.

Veg loss 2: Jan 2024



Veg loss 2: Dec 2024

Veg Loss 2 – Significant vegetation loss observed in area circled in yellow. This is most likely due to human or natural interference.

Veg loss 3: Jan 2024

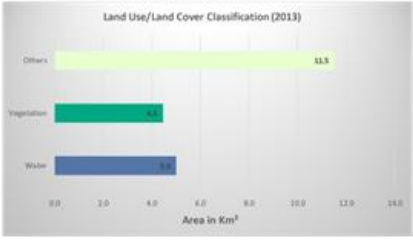
Veg loss 3: Dec 2024

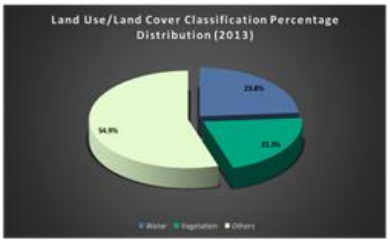
Veg Loss 3 – Pocket of vegetation loss observed in area circled in yellow. This is most likely due to human or natural interference.

BODO CLASSIFICATION OUTPUT AND STATISTICAL ANALYSIS (2013 – 2025)

2013

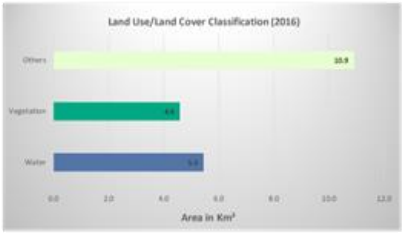
Epoch: 2013		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.0	23.8
Vegetation	4.5	21.3
Others	11.5	54.9
TOTAL	20.98	100

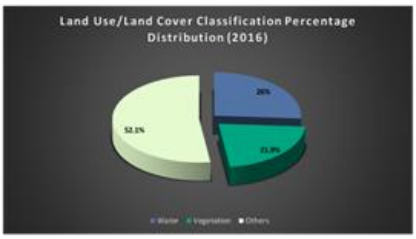




2016

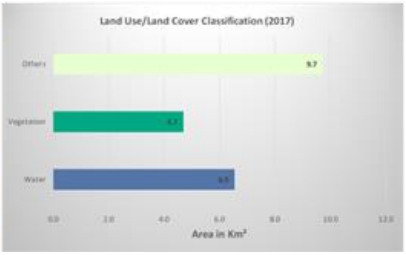
Epoch: 2016		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.5	26.0
Vegetation	4.6	21.9
Others	10.9	52.1
TOTAL	20.97	100

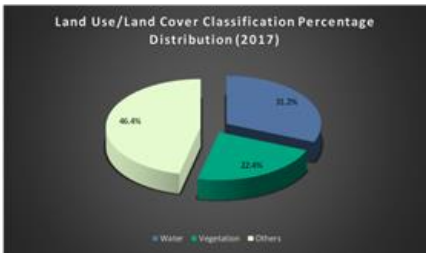




2017

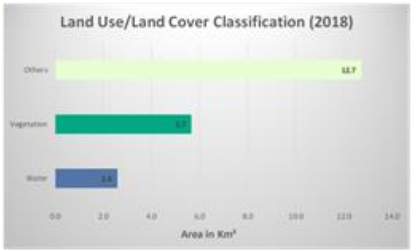
Epoch: 2017		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	6.5	31.2
Vegetation	4.7	22.4
Others	9.7	46.4
TOTAL	20.97	100

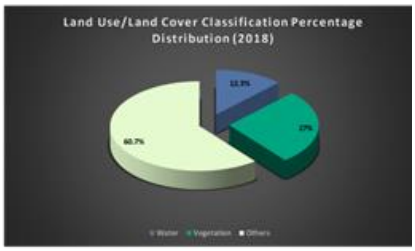




2018

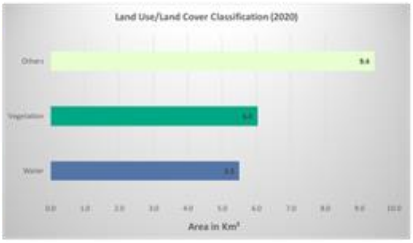
Epoch: 2018		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	2.6	12.3
Vegetation	5.7	27.0
Others	12.7	60.7
TOTAL	20.98	100

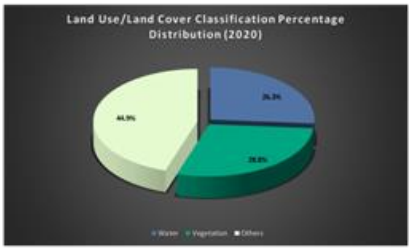




2020

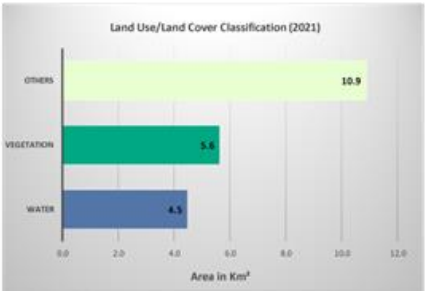
Epoch: 2020		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.5	26.3
Vegetation	6.0	28.8
Others	9.4	44.9
TOTAL	20.98	100

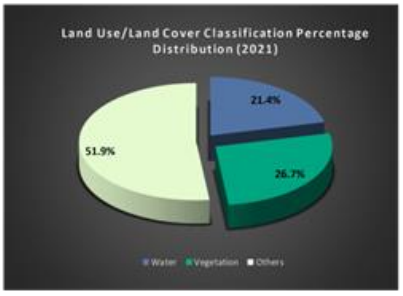




2021

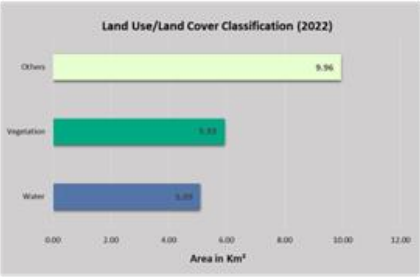
Epoch: 2021		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	4.5	21.4
Vegetation	5.6	26.7
Others	10.9	51.9
TOTAL	20.98	100

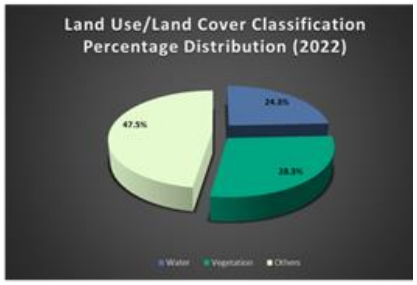




2022

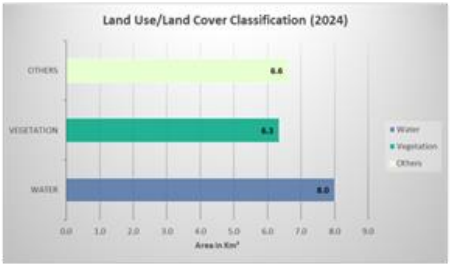
Epoch: 2022		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.09	24.3
Vegetation	5.93	28.3
Others	9.96	47.5
TOTAL	20.98	100

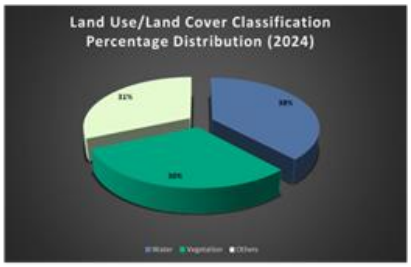




2024

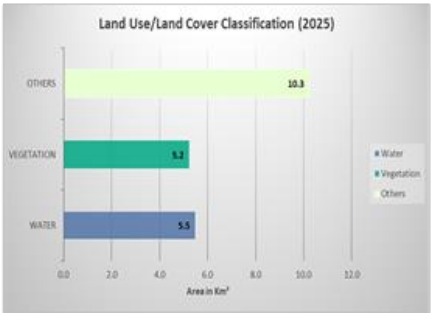
Epoch: 2024		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	8.0	38%
Vegetation	6.3	30%
Others	6.6	31%
TOTAL	20.98	100%

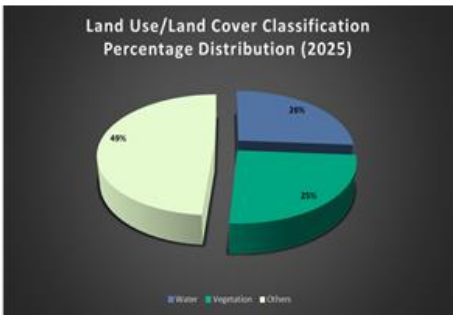


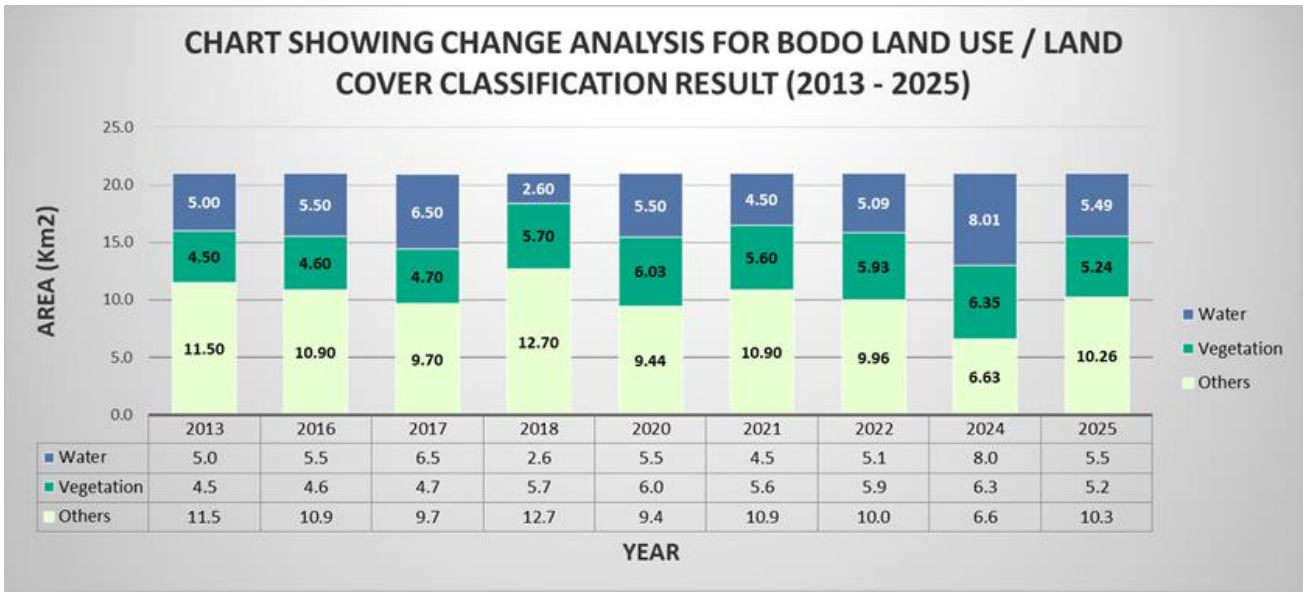


2025

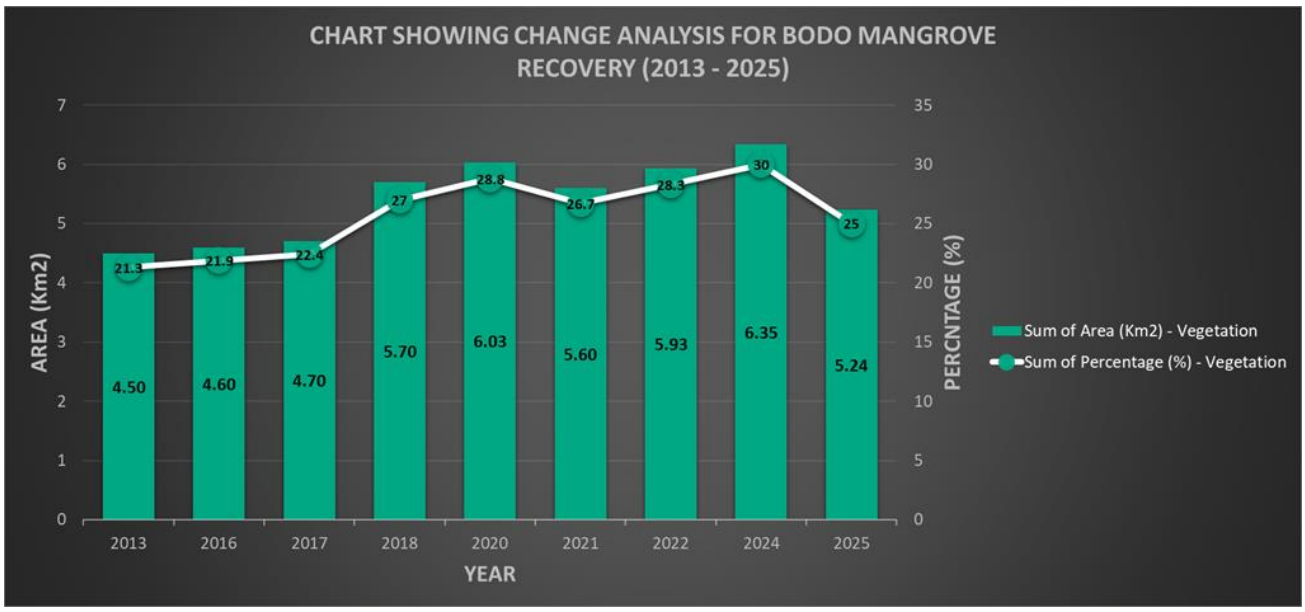
Epoch: Dec 2024 (For 2025 Study)		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.5	26%
Vegetation	5.2	25%
Others	10.3	49%
TOTAL	20.99	100%





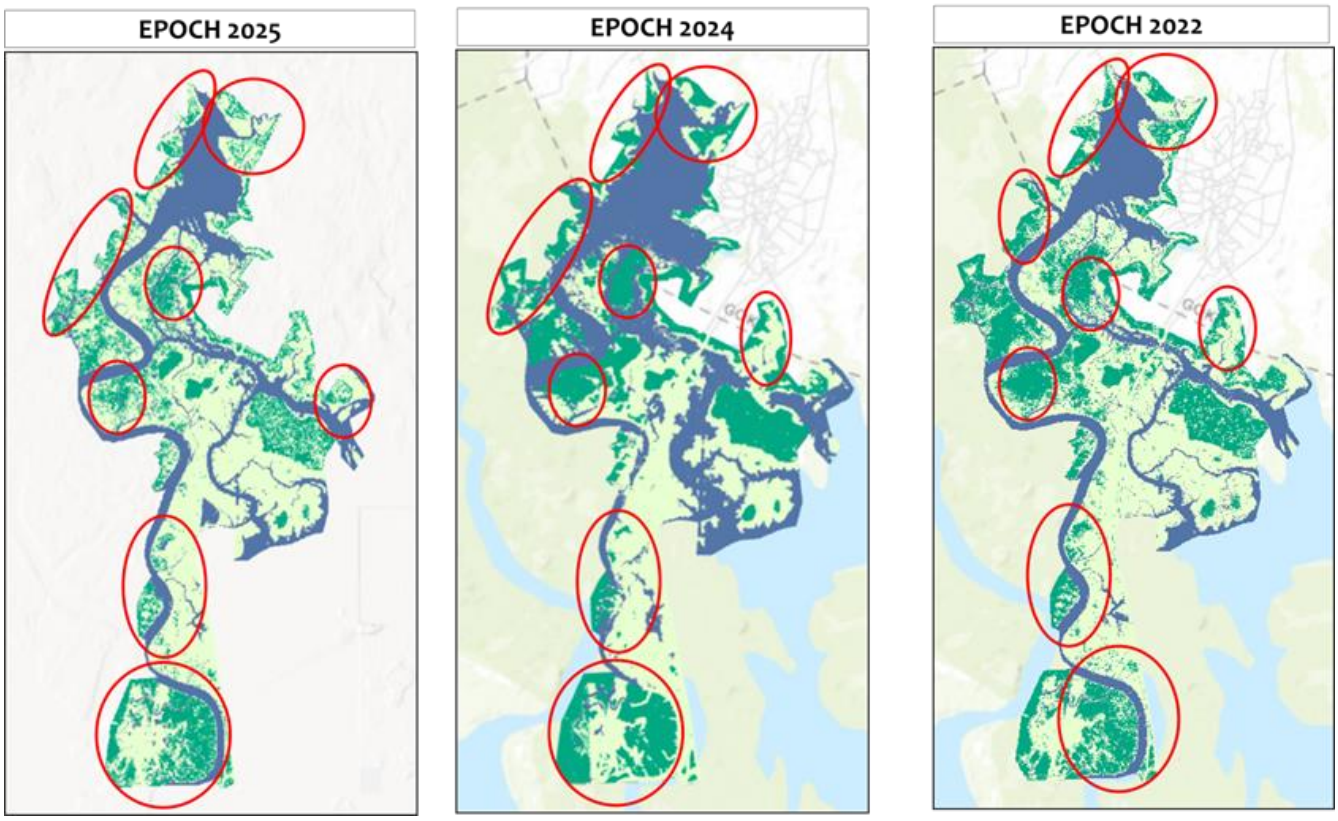


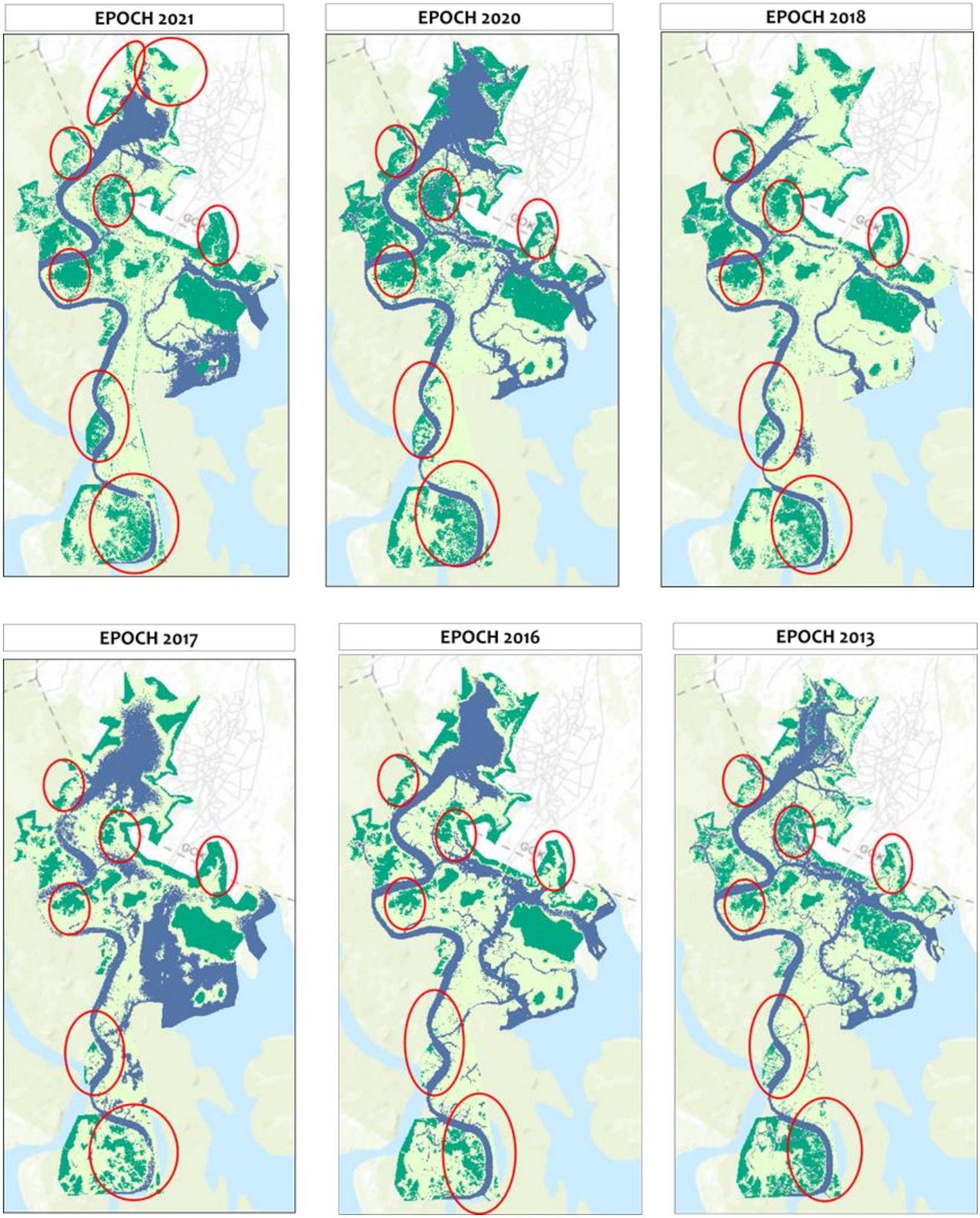
	2013	2016	2017	2018	2020	2021	2022	2024	2025
Water	5.0	5.5	6.5	2.6	5.5	4.5	5.1	8.0	5.5
Vegetation	4.5	4.6	4.7	5.7	6.0	5.6	5.9	6.3	5.2
Others	11.5	10.9	9.7	12.7	9.4	10.9	10.0	6.6	10.3



Charts showing change analysis for land use/land cover studies.

CHANGE ANALYSIS FOR CLASSIFICATION RESULT WITHIN BODO AOI





Bodo classification output and statistical analysis (2021 – 2025)

Year	Land Use/Land Cover Types	Area (Km2)	Percentage (%)
2013	Water	5.0	23.8
	Vegetation	4.5	21.3
	Others	11.5	54.9
2016	Water	5.5	26.0
	Vegetation	4.6	21.9
	Others	10.9	52.1
2017	Water	6.5	31.2
	Vegetation	4.7	22.4
	Others	9.7	46.4
2018	Water	2.6	12.3
	Vegetation	5.7	27.0
	Others	12.7	60.7
2020	Water	5.5	26.3
	Vegetation	6.0	28.8
	Others	9.4	44.9
2021	Water	4.5	21.4
	Vegetation	5.6	26.7
	Others	10.9	51.9
2022	Water	5.09	24.3
	Vegetation	5.93	28.3
	Others	9.96	47.5
2024	Water	8.01	38.0
	Vegetation	6.35	30.0
	Others	6.63	31.0
2025	Water	5.49	26.0
	Vegetation	5.24	25.0
	Others	10.26	49.0

Area and Percentage of The Land Use / Land Cover Classes Within Bodo AOI From 2013 – 2025

LULC	2024 AREA (sqr km)	2025 AREA (sqr km)	AREA CHANGED (sqr km)	STATUS
Water	8.01	5.49	2.52	Loss
Vegetation	6.35	5.24	1.11	Loss
Others	6.63	10.26	3.63	Gain
TOTAL	20.99	20.99		

Land Use / Land Cover Net Gain/Loss Change Between 2024 – 2025

CONCLUSIONS & RECOMMENDATIONS:

CONCLUSIONS:

The study successfully deployed remote sensing and geospatial technologies to map, extract and estimate area of vegetation cover, water, and others which was subsequently used for a time-lapse change analysis for mangrove regrowth in the area of study between year 2021 and year 2025, enabling the Ogoni Restoration Project (ORP) team monitor, plan and manage their replanting campaigns and other related activities.

The result of the statistical analysis for the 2025 classification study shows a land cover/land use percentage area distribution:

Water:

The water class reduced from 8.0 km² in January 2024 to 5.5 km² in December 2024 (for 2025 study year). This reduction in water level may be attributed to the changes in tide and water levels as at the time of image acquisition.

Vegetation

The vegetation class reduced from 6.3 km² in January 2024 to 5.2 km² in December 2024 (for 2025 study year). This is an approximate reduction of 1.1 km². This reduction is probably due to noticeable pockets and large mangrove vegetation loss distributed across the Bodo area of interest (see snapshots in page 15). From visual interpretation, the vegetation loss may be attributed to environmental/natural phenomena or human activities and interference. Judging from the image interpretation and analysis, the mangrove regrowth process by the ORP team appears to give a false progressive representation due to the significant vegetation loss observed across the entire Bodo AOI. We propose that all the Cell locations and the entire Bodo study area be given consideration for new replanting of mangrove seedlings most especially in areas where there are defoliation of vegetation and as well locations that the vegetation have completely disappeared as highlighted in the report (Pages 9 – 13).

Others:

The others class increased from 6.6 km² in January 2024 to 10.3 km² in December 2024 (for 2025 study year). The reason for the increase is probably attributed to changes in the water levels due to low tide showing extended riverbank covering most of the mud flats area and pockets of areas with significant vegetation loss.

The result of this analysis is to feed into a more robust analysis of all available satellite imagery over the Bodo remediation area of interest till date to have a clear view of the time lapse in mangrove distribution using Support Vector Machine classification method in combination with a segmented image, adequate training data sets and field verification of results.

RECOMMENDATIONS:

1. The actual status of vegetation, health and regrowth should be verified during field visits.
2. Locations of areas where subsequent replanting will be done should be shared by the ORP team to enable more focused monitoring and better analysis and interpretation of Remote sensing data.

*A story map is presently being created and can be accessed via this Ataba, Bodo Reforestation Project link and will be live on the 20th of October, 2025.

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